

Covariant Formulation of Shock Waves Propagation for Traffic Flow on Curved Roads

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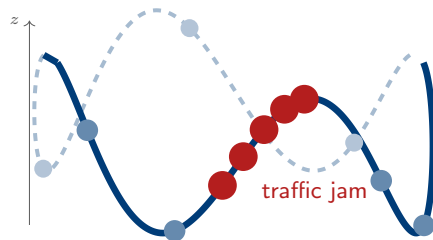
SIAM Conference, May 2026

Traffic jams on curved roads

Traffic waves propagate on **real road geometry**

— curves, hills, ramps —
but models assume **flat space**.

- Empirical evidence: slower flow at sharp turns, jam accumulation on inclines
- Highway Capacity Manual adjusts capacity for curvature — but *ad hoc*
- No first-principles coupling of geometry to shock dynamics



3D road with spatially varying geometry.
Vehicles cluster where curvature is high.

Phantom jams: no bottleneck needed

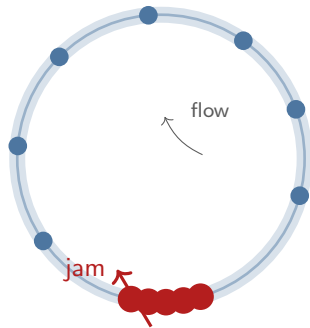
Sugiyama *et al.* (2008) [1]:

22 cars on a circular track. Uniform initial speed. **No traffic lights. No accidents.**

Within minutes: a **spontaneous traffic jam** propagating *backward* against the flow.

Even on a **flat circle** the system is unstable to density waves.

Question: What happens on a road with **spatially varying geometry**?



What exists:

- Conservation laws on Riemannian manifolds: well-posedness, FV schemes [2,3]
- Traffic on curved roads: lattice and macro models with curvature/slope effects [4,5,6]
- KdV–Burgers on curved manifolds: Laplace–Beltrami dissipation, shock profiles [7]
- Embedding methods for conservation laws on manifolds in $\mathbb{R}^3$ [8]

What is missing:

The gap

No published model combines:

- 1 A macroscopic traffic model (LWR)
- 2 on a spatial curve $\gamma(s) \in \mathbb{R}^3$
- 3 with **explicit** $\kappa(s), \tau(s)$ in the fundamental diagram
- 4 and derives how geometry modifies the **shock velocity**

This is what we do.

Spatial curves in $\mathbb{R}^3$: the Frenet–Serret frame

A road is a smooth closed curve $\gamma(s) \in \mathbb{R}^3$ parametrised by arc length s .

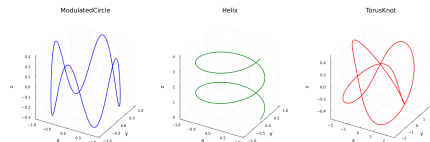
The **Frenet–Serret apparatus**:

$$\mathbf{T}'(s) = \kappa(s) \mathbf{N}(s)$$

$$\mathbf{N}'(s) = -\kappa(s) \mathbf{T}(s) + \tau(s) \mathbf{B}(s)$$

- $\kappa(s)$: **curvature** (how sharply the road turns)
- $\tau(s)$: **torsion** (how the road twists out of plane)

These are the **intrinsic geometric data** that should couple to flow dynamics.



Four test geometries: straight line

($\kappa = 0, \tau = 0$), circle ($\kappa = \text{const}, \tau = 0$),
helix ($\kappa, \tau = \text{const}$), modulated 3D curve
(κ varies $\sim 800\times$).

Covariant scalar conservation law

The Lighthill–Whitham–Richards (LWR) equation on a **curved road**:

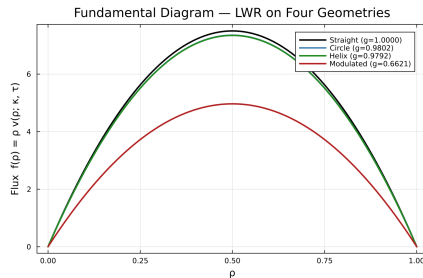
Covariant LWR equation

$$\partial_t \rho + \partial_s [\rho v(\rho; \kappa, \tau)] = 0$$

The flux depends on geometry via the **fundamental diagram**:

$$v(\rho; \kappa, \tau) = v_{\max} g(\kappa, \tau) \left(1 - \frac{\rho}{\rho_{\max}}\right)$$

with the **geometric correction**:



Fundamental diagrams (LWR) for the four geometries. Curvature reduces max flux by 2.0%; torsion adds 0.07%.

Rankine–Hugoniot conditions on curved geometry

At a shock discontinuity, the jump condition gives the **shock velocity**:

Geometric Rankine–Hugoniot

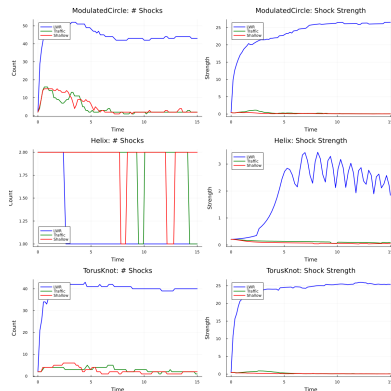
$$V_{\text{shock}}(s) = \frac{f(\rho_R; s) - f(\rho_L; s)}{\rho_R - \rho_L}$$

where $f(\rho; s) = \rho v(\rho; \kappa(s), \tau(s))$

For **constant geometry** (κ, τ uniform):

- $V_{\text{shock}} = v_{\text{max}} g(1 - \rho_L - \rho_R)$
- Exact prediction: no free parameters

For **variable geometry**: $V_{\text{shock}}(s)$ depends on position — the shock accelerates and decelerates as it traverses the road.



Shock analysis across geometries. For constant κ, τ : theory matches numerics to **machine precision**.

The effective potential: why $\partial_s \kappa$ matters

Expanding $\partial_s f$ in the conservation law:

$$\partial_t \rho + \lambda(\rho, s) \partial_s \rho = \underbrace{-\rho v(\rho) \partial_s \Phi}_{\text{geometric source}}$$

Effective potential

$$\Phi(s) = \beta_\kappa |\kappa(s)| + \beta_\tau |\tau(s)|$$

Physical interpretation:

- $\partial_s \Phi > 0$ (increasing curvature): density **accumulates**
- $\partial_s \Phi < 0$ (decreasing curvature): density depletes
- $\partial_s \Phi = 0$ (constant geometry): **no geometric forcing**

This explains **three key observations**:

- 1 **Helix** ($\partial_s \Phi = 0$): no jammitons
- 2 **Modulated curve** ($\partial_s \Phi$ large): density traps in high- κ regions
- 3 Shocks **refract** across Φ -gradients — analogous to Snell's law

We solve the covariant LWR equation with a **Godunov scheme** on the arc-length coordinate s :

- 1 Piecewise-constant reconstruction on each cell
- 2 Exact Riemann solver at each interface:
 $f^* = f(\rho^*)$ with ρ^* from the local Rankine–Hugoniot condition
- 3 Conservative update:

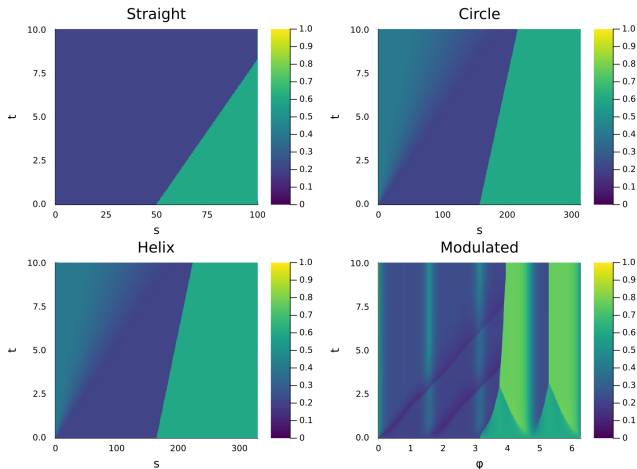
$$\rho_j^{n+1} = \rho_j^n - \frac{\Delta t}{\Delta s} (f_{j+1/2}^* - f_{j-1/2}^*)$$

Implementation:

- Julia package, ~ 2000 lines
- $N_s = 500$ spatial cells
- CFL: $\Delta t \leq \Delta s / \max |f'|$
- Periodic BCs
- Riemann solver uses local $\kappa(s_j), \tau(s_j)$

Also implemented: MUSCL reconstruction with TVD limiters (minmod, van Leer, MC) for second-order accuracy.

Kymographs: shock propagation across geometries



Space-time density maps $\rho(s, t)$ for the four test geometries. Riemann problem: $\rho_L = 0.2 \rightarrow \rho_R = 0.6$.

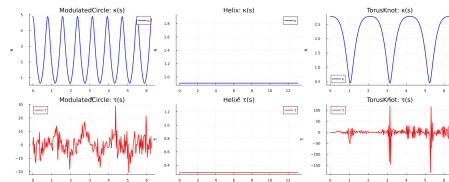
Constant geometry: uniform shock velocity. **Variable geometry:** shock speed modulated by local $\kappa(s)$.

Constant-geometry validation

For the three constant-geometry cases
($v_{\max} = 30$, $\beta_{\kappa} = 1$, $\beta_{\tau} = 0.5$):

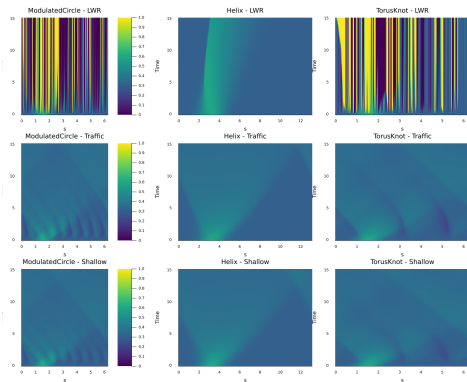
Geometry	κ	τ	V_{shock}
Straight	0	0	6.000 (ref.)
Circle ($R=50$)	0.02	0	5.881
Helix	0.02	0.002	5.877

- Numerical V_{shock} matches Rankine–Hugoniot to **machine precision** ($< 10^{-15}$)
- Curvature alone: **2.0%** slowdown
- Adding torsion: additional **0.07%**
- $q \equiv e^{-\beta_{\kappa}|\kappa| - \beta_{\tau}|\tau|}$ is **exact** for constant



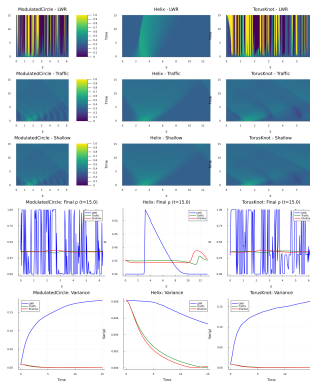
$\kappa(s)$ and $\tau(s)$ profiles. The modulated curve has κ varying by $\sim 800\times$.

Variable geometry: geometric refraction



Kymographs: shock front **bends** across regions of different κ .

Modulated circle: $\kappa \in [0.05, 40]$, velocity field $v \in [0, 13.6]$ — **1700% variation**. Shocks **nearly halt** at high κ and **accelerate** through low- κ stretches. This is the **geometric refraction** predicted by the effective potential $\Phi(s)$.

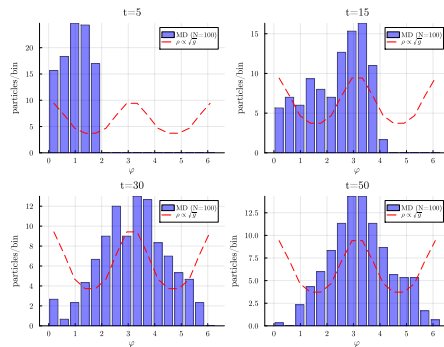
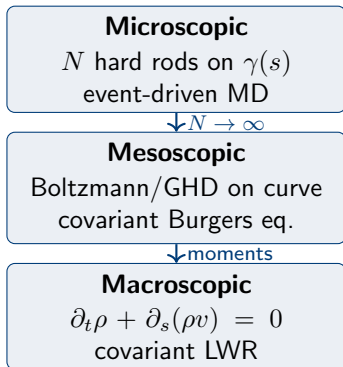


Density evolution,

shock trajectories, and velocity modulation on the modulated curve.

Connection to microscopic dynamics

The covariant LWR sits atop a **three-scale hierarchy** that we have verified independently:



Lines: covariant Burgers PDE. **Symbols:** molecular dynamics, 500 hard rods on an ellipse. **24 initial conditions**, no free parameters.

Elastic hard rods **do not** form Burgers shocks — shocks require the macroscopic

Key theoretical property:

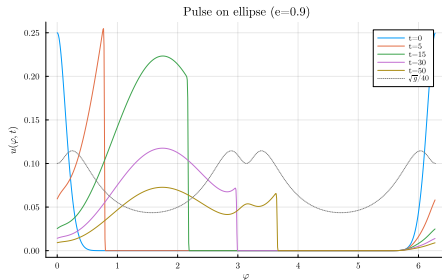
In arc-length coordinates s , the covariant equation reduces to **flat-space LWR** with a position-dependent fundamental diagram.

All exact solutions of flat Burgers *transplant* to the curved road:

- N-wave $\rightarrow$ metric-distorted N-wave
- Sawtooth $\rightarrow$ metric-modulated sawtooth
- Cole–Hopf viscous profile $\rightarrow$ curved Cole–Hopf

Consequence

Shock dynamics on *any* geometry = flat-space theory + local $q(\kappa, \tau)$.



N-wave on an ellipse: the two fronts travel at different speeds — the pulse **distorts asymmetrically**.

What we showed:

- Covariant LWR on curves in $\mathbb{R}^3$ with $\kappa(s), \tau(s)$ in the flux
- Effective potential $\Phi(s)$ governs density redistribution
- Godunov: mass error $< 10^{-15}$
- Constant geometry: **exact** V_{shock}
- Variable geometry: **geometric refraction**

Future work:

- Agent-based IDM validation on curves
- GPS data on curved highways
- Active matter on manifolds
- Higher-order MUSCL/WENO schemes
- Networks of curved roads

Geometry	V_{shock}	Error vs. R-H	Effect
Straight ($\kappa=0$)	6.000	0 (ref.)	—
Circle ($\kappa=\text{const}$)	5.881	$< 10^{-15}$	-2.0%
Helix ($\kappa, \tau=\text{const}$)	5.877	$< 10^{-15}$	-2.07%
Modulated (κ variable)	varies	pos.-dependent	geom. refraction

Thank you!

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Questions?

- [1] Sugiyama *et al.*, New J. Phys. **10**, 033001 (2008)
- [2] Ben-Artzi & LeFloch, Ann. IHP Anal. Non Lin. **24**, 989 (2006)
- [3] LeFloch, Okutmustur & Neves, Acta Math. Sinica **25**, 1041 (2008)
- [4] Zhou & Shi, Nonlinear Dyn. **83**, 1217 (2016)
- [5] Xue *et al.*, Nonlinear Dyn. **95**, 3295 (2019)
- [6] Tian & Kang, Symmetry **17**, 1299 (2025)
- [7] Palencia, ZAMM **106**, e70435 (2026)
- [8] Wang *et al.*, J. Sci. Comput. **93**, 56 (2022)
- [9] El & Hofer, SIAM Rev. **59**, 3 (2017)
- [10] Lighthill & Whitham, Proc. R. Soc. A **229**, 317 (1955)

Backup: Simulation parameters

The geometric correction to the fundamental diagram accounts for the reduced effective speed on curves:

A vehicle on a curve of curvature κ experiences centripetal acceleration $a_c = v^2 \kappa$. Stability requires $a_c \leq \mu g_{\text{grav}}$, giving the speed reduction:

$$v_{\max}(\kappa) = v_{\max}^{\text{flat}} \sqrt{\frac{\mu g_{\text{grav}}}{\kappa}} \quad \text{for large } \kappa$$

The exponential model $g = e^{-\beta_\kappa |\kappa|}$ interpolates smoothly between flat ($\kappa = 0$) and highly curved regimes.

Parameters used:

- $v_{\max} = 30.0$
- $\rho_{\max} = 1.0$
- $\beta_\kappa = 1.0$ (curvature sensitivity)
- $\beta_\tau = 0.5$ (torsion sensitivity)
- $\alpha = 1$ (Greenshields)
- Riemann: $\rho_L = 0.2$, $\rho_R = 0.6$

Test geometries:

- Circle: $R = 50$, $\kappa = 1/R$,
 $\tau = 0$
- Helix: $R = 50$, $h = 100$
- Modulated: $\gamma(\varphi) =$
 $(a \cos \varphi, a \sin \varphi, a^2 \cos 2\varphi),$

Backup: Covariant Burgers equation

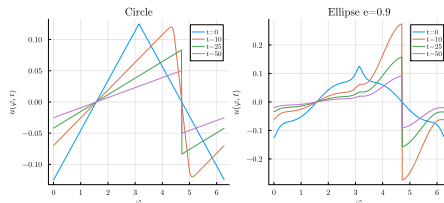
From the kinetic hierarchy (moment closure), the velocity field satisfies the **covariant Burgers equation**:

$$\partial_t u + u \partial_\varphi u + \Gamma(\varphi) u^2 = 0$$

where $\Gamma = g'/(2g)$ is the Christoffel symbol of the metric $g(\varphi) = |\gamma'(\varphi)|^2$.

Key property: in arc-length coordinates s , the physical velocity $v = \sqrt{g} u$ satisfies the flat Burgers equation.

This establishes **exact equivalence** between curved and flat dynamics up to coordinate transformation.



N-wave distortion on the ellipse. The

Christoffel term Γu^2 accelerates the wave in low- g regions and decelerates it in high- g regions.